POLYLIQUID

Decentralized Delegated Reputation Oracle

March 2026

Money can be bought instantly. Trust cannot.

1. Executive Summary

Polyliquid is a Decentralized Delegated Reputation Oracle designed to replace existing oracle systems for prediction markets. The protocol combines delegated staking with reputation-based voting to create an oracle where the cost of attack **always exceeds the market size** - 219% for markets under $10M and 110% for larger markets.

The dispute layer uses **dual DAO arbitration**: both the Polyliquid DAO and the client's DAO (e.g. Polymarket) must vote in parallel to confirm or overturn a result. An attacker would need to bribe both DAOs simultaneously - an economically impossible task.

Core innovation: Voting weight = LP_stake × Reputation, where reputation is earned through months of correct voting and cannot be purchased. The system self-balances: larger markets generate more fees, attracting more LP, which increases the cost of attack proportionally.

1.1 Key Metrics

Metric	Value
Oracle fee	**0.1% of resolved OI**
LP target APY	50% (market-driven)
Attack cost R1 (< $10M)	**219% of market size**
Attack cost R1 (> $10M)	**110% of market size**
Attack cost R1 + Dual DAO	**∞ (two DAOs reverse independently)**
UMA attack cost	**$2.5M fixed (5M tokens)**
Resolution time (< $10M)	8 hours
Resolution time (> $10M)	16 hours
Weight formula	W = Stake × Rep (κ = 1.0)
LP slash (R1)	20% flat for all tiers
LP slash (Dispute)	**100% flat (total destruction)**
Protocol share (π)	10% of fees → Foundation
Polymarket resolved OI	**$2.5B/month (Jan–Feb 2026)**
Day-1 fee revenue	**$2.5M/month = $30M/year**
Day-1 Foundation revenue	**$250K/month = $3M/year**

2. The Problem: Oracle Vulnerability

Prediction markets require oracles to determine event outcomes. For subjective events (politics, economics, culture), automated oracles like Chainlink cannot be used. Human-powered oracles are required.

The dominant solution today is UMA (Universal Market Access), which uses token-weighted voting. UMA's fundamental flaw: voting power equals token holdings. Influence can be purchased instantly.

2.1 UMA Failure Cases

March 2025: $7M stolen. A whale acquired 5M UMA tokens (~$2.5M) representing 25% of voting power, and forced a false resolution on the Trump/Ukraine minerals deal. Polymarket confirmed the error but refused refunds. After the attack, the whale retained the tokens - they were not slashed.

July 2025: $242M dispute. Market: "Will Zelenskyy wear a suit?" $242M volume. 40+ media outlets reported "Yes." UMA voted "No." Top 10 UMA whales controlled 30% of votes. Polymarket began seeking a replacement oracle. October 2025: POLY token announced.

Root cause: UMA attack cost = 5M tokens (~$2.5M) + $750 in transaction fees. This is a fixed amount regardless of market size - the same $2.5M attacks a $1M market and a $242M market. After the attack, tokens are not slashed and can be sold. UMA's response: a whitelist of 37 addresses = centralization.

2.2 Market Size & Opportunity

In January and February 2026, Polymarket resolved **$2.5 billion per month** through UMA oracle. Extrapolating conservatively to a full year gives **$30 billion in annual resolved OI** that could be serviced by Polyliquid in 2026.

Accounting for market growth (prediction markets grew 3–5× annually in 2024–2025) and expansion to other prediction market platforms beyond Polymarket, the realistic addressable market reaches **$100 billion per year** in resolved OI requiring human oracle services.

At Polyliquid's 0.1% commission:

Scenario	Annual Resolved OI	Annual Oracle Fees
Conservative (Polymarket only)	**$30B**	**$30M**
Realistic (all platforms + growth)	**$100B**	**$100M**

Of the fee revenue, 90% goes to voters and LP providers as rewards, and 10% goes to the Polyliquid Foundation.

3. Protocol Design

3.1 Core Mechanism: Three Layers of Protection

Layer 1: LP → Voter Delegation. Liquidity Providers choose voters based on track record and delegate capital. Voters stake this capital on market outcomes. Commission rates range from 0–99%, set by the market based on liquidity supply and market resolution volumes.

Layer 2: Commit-Reveal Voting with LP Exit. Voters submit encrypted votes (commit phase), then reveal them. During the Reveal phase, LP providers can see how their voter voted and withdraw their liquidity if they disagree. This is the collusion detection mechanism.

Layer 3: Dual DAO Dispute Resolution. Anyone can dispute a Round 1 result by posting a bond equal to 2% of market MaxPayout. Two DAOs vote simultaneously over 48 hours: the Polyliquid DAO (PL token holders) and the client platform's DAO (e.g. Polymarket's POLY token holders). Both DAOs must reach quorum and vote independently. See Section 6 for detailed mechanics.

3.2 Reputation Formula

```
R(t) = Σ LP_stake × 0.995^(days since vote)
```

Each correct vote adds the LP stake amount to the voter's reputation. Old votes decay at 0.5% per day, giving a half-life of 138 days.

```
Reputation ceiling = daily_stake / 0.005
```

For an elite voter staking $1M per day, the maximum achievable reputation is $200M. This ceiling is reached asymptotically after 12–18 months of consistent voting.

3.3 Why Flat Decay (Not Progressive)

Progressive decay (faster decay for higher reputation) creates a Sybil vulnerability: splitting $200M reputation into 2×$100M reduces total decay. **Flat δ = 0.995** ensures splitting gives zero advantage. Combined with κ = 1.0, the system is fully Sybil-neutral.

3.4 Voting Weight: κ = 1.0 (Sybil-Neutral)

```
W = LP_stake × Rep^1.0
```

The exponent κ = 1.0 is used for both voting weight AND reward distribution. One account with $200M reputation receives exactly the same rewards as two accounts with $100M each. Sybil attacks are economically neutral.

3.5 LP Exit During Reveal Phase

A critical innovation: during the Reveal phase, LP providers can see how their voter voted and choose to withdraw their liquidity before finalization. The Reveal window is 2 hours for markets under $10M and 6 hours for larger markets.

If LP stays: LP shares the voter's fate. If the voter is wrong, LP's stake is slashed. This implies LP either agrees with the vote or is in collusion.

If LP exits: LP's stake is removed before finalization. LP is not slashed. The voter is slashed on remaining stake only.

Situation	LP Action	LP Result	Voter Result
Voter correct	Stayed	**Reward**	**Reward**
Voter wrong, LP honest	Exited in Reveal	**0% slash**	**20% slash**
Voter wrong, LP stayed	Stayed	**20% slash**	**20% slash**
Attack + Dispute	Stayed (collusion)	**100% slash**	**100% + Rep=0**

4. Slashing Mechanisms

4.1 Flat LP Slash: 20% (R1) / 100% (Dispute)

Scenario	LP Slash Rate	Rationale
R1 (no dispute)	20% flat	Standard penalty for incorrect vote
R1 + Dual DAO overturns	**100% flat**	Total loss. LP had exit window.
Reputation (R1)	Up to 50%	Proportional to consensus margin
Reputation (Dispute)	**100% (reset to 0)**	Complete destruction of trust

Why flat slash: LP faces the same risk regardless of voter tier. Combined with κ = 1.0, newcomers can attract LP by offering competitive commission rates on a level playing field.

4.2 Reputation Slashing

```
S_rep = S_base × (W_majority − 50%) / 50%
```

Close vote (51/49) = 1% slash. Clear consensus (80/20) = 30% slash. Dual DAO dispute: 100% reputation reset to zero.

4.3 Cascade of Destruction

When a voter participates in an attack overturned by Dual DAO:

Step 1: **Both DAOs overturn R1** → attacker's reputation slashed to zero.

Step 2: **100% LP slash** → LP loses all capital (they stayed through Reveal = collusion).

Step 3: **Rep = 0** → no LP will delegate to this voter again.

Step 4: **No LP = no stake = no weight = no rewards = no income.**

Step 5: **Recovery takes ~12 months** of farming with own capital.

5. Attack Cost: Self-Balancing Security

5.1 The Core Insight

Previous oracle models calculated attack cost from average locked liquidity. Polyliquid's security is **derived from the fee structure itself**, creating a mathematical guarantee that scales automatically.

Key parameters: 0.1% commission from resolved OI, 50% APY target for LP providers, 100% slash on dispute. Resolution time: 8 hours (<$10M) or 16 hours (>$10M).

5.2 Derivation: Markets < $10M (8 hours)

Step 1: Fee from market. A market with resolved OI = V generates a fee of V × 0.1%.

$$Fee = V \times 0.001$$

Step 2: LP required. For LP to earn 50% APY, the fee must equal the LP's return for 8 hours of lockup:

$$Fee = LP \times APY \times (8 / 8760)$$

Solving for LP:

$$LP = V \times 0.001 / (0.50 \times 8/8760) = V \times 0.001 / 0.000457 = V \times 2.19$$

The market is secured by 2.19× its own size in LP capital.

Step 3: Attack cost (100% slash). If Dual DAO overturns the vote, all LP that stayed through Reveal is slashed 100%:

$$Attack_cost = LP \times 100\% = V \times 2.19 = 219\% \text{ of market}$$

5.3 Derivation: Markets > $10M (16 hours)

Larger markets use a 16-hour resolution cycle (2h Commit + 6h Reveal + 8h Dispute). Extended Reveal and Dispute windows provide additional protection.

$$LP = V \times 0.001 / (0.50 \times 16/8760) = V \times 0.001 / 0.000913 = V \times 1.095$$

$$Attack_cost = V \times 1.095 \times 100\% = 110\% \text{ of market}$$

At 110%, the attack is unprofitable on LP slash alone. Adding the dispute bond (2%) brings the minimum to **112%**. NPV of lost future income adds further cost.

5.4 With Dual DAO: Infinite Protection

$$Attack_R1{+}DualDAO = \infty$$

The Dual DAO system can overturn any R1 result. Both the Polyliquid DAO and the client's DAO vote simultaneously. Attacker must bribe both DAOs - see Section 6 for why this is impossible.

5.5 Self-Balancing Security Table

Market	Tier	Time	Fee (0.1%)	LP locked	Slash 100%	Attack
$1M	< $10M	8h	$1K	$2.19M	**$2.19M**	**219%**
$10M	< $10M	8h	$10K	$21.9M	**$21.9M**	**219%**
$50M	> $10M	16h	$50K	$54.8M	**$54.8M**	**110%**
$100M	> $10M	16h	$100K	$109.5M	**$109.5M**	**110%**
$1B	> $10M	16h	$1M	$1.095B	**$1.095B**	**110%**

6. Dual DAO Governance & Dispute Resolution

6.1 Why Dual DAO

Round 1 (voter consensus) provides >100% attack cost under equilibrium conditions. However, during early growth phases or for extremely large markets, actual LP may be less than the formula requires. The Dual DAO exists as an absolute guarantee: regardless of LP levels, the correct outcome can always be restored.

Unlike single-DAO systems, Polyliquid uses **two independent DAOs** that vote in parallel. An attacker must corrupt **both** organizations simultaneously to override a dispute - a requirement that makes the attack economically impossible at any scale.

6.2 The Two DAOs

	Polyliquid DAO	Client DAO
Token	PL (Polyliquid governance token)	Client token (e.g. POLY)
Voters	PL token holders	Client token holders
Quorum	Set by Polyliquid governance	Set by client platform
Motivation	Protect protocol reputation and LP	Protect platform credibility and users
Independence	**Completely separate token base**	**Completely separate token base**

6.3 How Dual DAO Voting Works

Trigger: Any participant who disagrees with the R1 result posts a bond equal to 2% of market MaxPayout.

Bond examples: $1M market → $20K bond. $10M → $200K. $100M → $2M. $1B → $20M.

Parallel voting: Both DAOs vote simultaneously over a 48-hour window. Each DAO votes according to its own governance settings and quorum requirements. Both DAOs are financially motivated to vote correctly - incorrect outcomes damage their platforms, token value, and user trust.

Decision logic: If both DAOs agree to overturn: R1 result is reversed, attackers slashed 100%. If both DAOs confirm R1: result stands, disputer loses bond. If DAOs disagree: the dispute enters an extended resolution period with additional scrutiny.

Progressive decentralization: At launch, Polyliquid founders hold a meaningful share of PL tokens and act as honest backstop. As the token distributes through airdrops, incentives, and trading, governance becomes fully decentralized. The client DAO is already decentralized (e.g. Polymarket with thousands of POLY holders).

6.4 Vote Outcomes & Distribution

Scenario	R1 Slashes	Bond	Client Gets	DAO PL Gets

No dispute	→ correct voters	-	$0	$0
Both DAOs confirm R1 (spam)	75% → correct voters	→ client	**100% of bond**	**25% of slashes**
Both DAOs overturn (attack)	50% correct, 25% client, 25% PL	Returned	**25% of slashes**	**25% of slashes**

6.5 Why Dual DAO Cannot Be Bribed

Two independent organizations. The Polyliquid DAO and the client DAO (e.g. Polymarket) have completely separate token bases, separate governance structures, and separate financial incentives. An attacker cannot influence both through a single action.

Client DAO scale. For a platform like Polymarket, the client DAO FDV is **$10B+**. Acquiring >50% voting power requires $5B+ in capital - far exceeding any possible attack profit.

Financial motivation. Both DAOs are financially motivated to vote honestly. The Polyliquid DAO protects the PL token value and LP provider trust. The client DAO protects platform credibility and user deposits. A corrupt vote destroys both.

Self-destructive economics. Any DAO voter who accepts a bribe to vote incorrectly destroys their platform's credibility, causing their own token holdings to collapse in value. The bribe must exceed not just the voting tokens, but the total portfolio loss - across both DAOs simultaneously.

On-chain transparency. All DAO votes are on-chain. Large token movements before a controversial vote are publicly visible across both platforms and would trigger community response before votes conclude.

7. Resolution Timeline

Phase	< $10M Market	> $10M Market	Purpose
Commit	2 hours	2 hours	Encrypted votes submitted
Reveal + LP Exit	2 hours	6 hours	LP sees vote, can withdraw
Dispute Window	4 hours	8 hours	Anyone posts 2% bond
Total R1	**8 hours**	**16 hours**	
+ Dual DAO (if disputed)	+ 48h = 56h total	+ 48h = 64h total	Two DAOs vote in parallel

99% of markets resolve within 8–16 hours without dispute.

8. Power Distribution

```
W = LP_stake × Rep^1.0    |    κ = 1.0 (Sybil-neutral)
```

Voting weight is the product of two independent factors: **capital (LP stake)** and **trust (reputation)**. Neither factor alone is sufficient.

Money alone is worthless. A whale with $10M LP stake but zero reputation has W = $10M × $0 = 0. No influence. This is the fundamental difference from UMA, where $10M buys instant control.

Reputation cannot be bought. Built over months of correct voting, decays daily. Even purchasing an account faces the decay clock - without continued correct voting, reputation erodes (half-life: 138 days).

9. Bootstrapping: Points & Airdrop

9.1 The Cold Start Problem

With κ = 1.0, elite voters dominate rewards proportional to reputation. Newcomers earn minimal rewards initially. LP naturally flows to elite voters.

9.2 Points System

Newcomers earn points proportional to their stake when farming with own capital: Points = Σ own_stake × days_active. Points continue until 100× capital turnover, $5M reputation, or 12 months elapsed.

9.3 Airdrop

Protocol allocates tokens for airdrop based on accumulated points. Speculative motivation for newcomers to farm even when direct rewards are minimal.

10. Economics

10.1 Revenue Model

Oracle commission: **0.1% of resolved OI**. This fee is paid by the prediction market platform (embedded in trader payouts), not by Polyliquid's clients directly.

10.2 Day-1 Revenue (Polymarket Replacement)

Polymarket resolved **$2.5 billion per month** through UMA in January and February 2026. If Polyliquid replaces UMA as the oracle:

Metric	Value
Monthly resolved OI	**$2.5B**

Fee (0.1%)	**$2.5M/month**
Voter + LP pool (90%)	$2.25M/month = $27M/year
Foundation (10%)	**$250K/month = $3M/year**
Annual fee total	**$30M/year**

10.3 Market Size

Scenario	Annual Resolved OI	Annual Oracle Fees
Conservative (Polymarket only, extrapolating Jan–Feb 2026)	**$30B**	**$30M**
Realistic (market growth + other prediction platforms)	**$100B**	**$100M**

10.4 Client Economics

Item	Value
Cost to client (e.g. Polymarket)	**$0 (fee embedded in trader payouts)**
Revenue on spam disputes	**100% of bond**
Revenue on real attacks	**25% of slashes**
Permissionless markets	Enabled (oracle scales automatically)
Reputational risk	Delegated to Polyliquid

11. Attack Resistance Analysis

11.1 Direct Voter Bribery

Attacker must bribe voters with >50% weight. With diverse voter base, minimum coalition: 25+ people. Even if bribed, LP exit during Reveal can reduce their weight. After dispute: 100% slash + rep = 0.

11.2 LP Exit as Defense

Honest LP exits during Reveal when they see suspicious votes. This reduces attacker weight, signals problems, protects LP capital, and leaves colluding LP to face 100% slash.

11.3 Sybil Attack

With κ = 1.0: zero advantage from splitting. One account $200M rep = two accounts $100M each.

11.4 Reputation Farming

Time to elite: 12–18 months of $1M daily stake. After attack: all destroyed, 12+ months to recover. NPV of elite career: $500K+/year at 50% APY. Economically irrational.

11.5 Dual DAO Corruption

See Section 6.5. Attacker must bribe both the Polyliquid DAO and the client DAO simultaneously. With client DAO FDV at $10B+ (e.g. Polymarket), acquiring >50% voting power in just one DAO exceeds any possible attack profit. Bribing both is categorically impossible.

11.6 Undercollateralized Markets (Early Stage)

During early growth, LP may be less than equilibrium requires. See Appendix A for a detailed $1B market case study. Key finding: even with 10× LP deficit, the Dual DAO makes the attack unprofitable (-$85M loss for attacker).

12. Protocol Parameters

Parameter	Value	Description
κ	**1.0**	Linear weight + reward exponent (Sybil-neutral)
δ	0.995/day	Reputation decay (half-life 138 days)
S_base	50%	Maximum reputation slash per vote
LP_slash_R1	**20% flat**	LP slash in R1 (all tiers equal)
LP_slash_dispute	**100% flat**	LP slash on Dual DAO overturn (total loss)
APY target	50%	LP provider target return (market-driven)

Commission	0.1%	Oracle fee on resolved OI
Protocol share π	10%	Foundation revenue from fees
Bond	2% MaxPayout	Dispute threshold for Dual DAO
Reveal window	2h / 6h	< $10M / > $10M (LP exit window)
Dispute window	4h / 8h	< $10M / > $10M markets
Dual DAO vote	48 hours (parallel)	PL DAO + Client DAO, independent quorum

13. Competitive Comparison

	UMA	Polyliquid
Attack cost	**$2.5M fixed**	**110–219% of market**
Cost scaling	Same for $1M and $1B	**Proportional to market**
Vote weight	Tokens ($)	**Stake × Rep (κ=1.0)**
Sybil resistance	None	**κ=1.0 (neutral)**
Collusion detection	None	**LP Exit in Reveal**
Slash on attack	0% (tokens kept)	**100% flat**
Dispute system	Single DAO (UMA token)	**Dual DAO (PL + Client)**
Resolution time	~48h	**8–16h (99%)**
2025 exploits	**$7M + $242M**	∞ with Dual DAO
Client revenue	$0	**Bonds + slashes**
Oracle fee	Funded by UMA inflation	**0.1% of resolved OI**

14. Roadmap

Phase 1: Wisdom Markets Integration (Q2–Q3 2026). First live deployment as the oracle for Wisdom Markets - a B2B prediction market platform for the iGaming industry, powered by SoftSwiss infrastructure. Planned resolved volume: $100M/month. Distribution: 400+ casinos, 6 million monthly active users. Oracle revenue at 0.1%: $100K/month.

Phase 2: Expansion to Other Prediction Markets (Q3–Q4 2026). Onboard additional prediction market platforms seeking a reliable oracle alternative. Grow voter community, build track record across diverse market types. Points program and early airdrop for voters.

Phase 3: Polymarket (2027). Replace UMA as the primary oracle for Polymarket. $2.5B+/month in resolved OI. Permissionless market creation enabled by self-balancing security. Attack cost 110–219% of market size vs UMA's fixed $2.5M. Full airdrop for early participants.

Appendix A: Attack Case Study - $1B Market

This appendix walks through a complete attack scenario on a $1 billion prediction market during Polyliquid's early growth phase, when LP is **significantly below equilibrium**.

A.1 Setup

Parameter	Value
Prediction market	Client platform (e.g. Polymarket)
Market question	"Will [impossible event] happen?"
Market OI	**$1,000,000,000**
YES price	$0.01 (market considers impossible)
Polyliquid fee (0.1%)	$1,000,000
Required LP (formula)	$1.095B (V × 1.095)
Actual LP on Polyliquid	**$100M (early stage - 10× deficit)**

A.2 Attacker's Position

Step 1: Attacker buys $10M worth of YES tokens at $0.01 on the prediction market.

Result: 1,000,000,000 YES shares. If resolved YES → payout = $1B. Profit = $990M.

A.3 Layer 1: Bribing R1 Voters

Attacker needs >50% of voting weight on Polyliquid to force a YES resolution.

Total LP staked: $100M across all voters.

Need to bribe: Voters controlling >$50M in delegated LP.

Each bribed voter risks:

• If no dispute: 20% LP slash = $10M total (on $50M).

• If dispute + Dual DAO overturn: 100% slash = $50M + reputation → 0 + 12 months lost income.

Minimum bribe cost: Attacker must compensate voters for worst case:

Component	Amount
Compensate 100% LP slash	**$50,000,000**
Compensate lost NPV (12 months at 50% APY)	~$25,000,000
Risk premium (coordination of 25+ voters)	~$5,000,000
Total R1 bribe cost	**~$75–80M**

R1 alone does NOT protect this market. Bribe cost ($75M) is only 7.5% of attacker profit ($990M). This is why the Dual DAO exists.

A.4 Layer 2: Dual DAO Dispute

An honest participant sees the incorrect YES resolution and posts a dispute bond:

Action	Detail
Dispute bond	2% × $1B = $20,000,000
Voting period	48 hours (both DAOs in parallel)
DAO 1: Polyliquid	PL token holders vote independently
DAO 2: Client (e.g. Polymarket)	POLY token holders vote independently
Required	**Both DAOs must agree to overturn**

Both DAOs vote to overturn. Result restored to NO (correct).

Consequences:

- Bribed R1 voters: **100% LP slash ($50M destroyed) + reputation = 0**
- Disputer: **$20M bond returned**
- Attacker: spent $75M on bribes + $10M on position = **-$85M total loss**
- Attacker's position on prediction market: resolved NO → **payout = $0**

A.5 Can the Attacker Bribe Both DAOs?

For the attack to succeed, the attacker must corrupt **both** DAOs simultaneously:

DAO	Token FDV	>50% Cost	Feasibility
Polyliquid DAO	Early stage	Varies	Potentially vulnerable alone
Client DAO (e.g. Polymarket)	**$10B+**	**>$5B**	**Impossible**
Both DAOs combined	**$10B+ combined**	**>$5B+**	**Impossible**

Even if the Polyliquid DAO were hypothetically vulnerable in its early stage, the **client DAO (e.g. Polymarket, FDV $10B+) is categorically unbribable**. The attacker's maximum profit of $990M is less than 20% of what would be needed to acquire >50% voting power in the client DAO alone.

The Dual DAO design ensures that the protocol inherits the security of its largest client's governance system.

A.6 Complete Attacker P&L

Scenario	Attacker Spends	Attacker Gets

R1 succeeds, no dispute	$10M position + $75M bribe = $85M	**$1B payout → +$915M profit**
R1 succeeds, Dual DAO overturns	$10M + $75M = $85M	**$0 payout → -$85M loss**
R1 + both DAOs bribed	$85M + $5B+ = $5.085B+	**$1B payout → -$4B+ loss**

Key insight: the only profitable scenario (R1 bribed, no dispute) requires that nobody in the entire ecosystem notices a $1B market resolved incorrectly and posts a $20M bond within the dispute window. For a market this size, this is effectively impossible - the incentive to dispute is enormous (25% of $50M+ in slashes).

A.7 Why Early-Stage LP Deficit is Acceptable

This case study demonstrates that even with 10× LP deficit ($100M vs required $1.095B):

• **R1 alone is insufficient** - bribe cost is only 7.5% of profit. This is an honest assessment.

• **Dual DAO makes the attack unprofitable** - attacker loses $85M, with zero chance of corrupting the client DAO.

• **The dispute incentive is enormous** - anyone can earn 25% of $50M+ in slashes by posting a $20M bond.

• **The protocol inherits client security** - even a young Polyliquid DAO is protected by the client's established governance.

• **As LP grows toward equilibrium, R1 alone becomes sufficient** - the Dual DAO transitions from essential backstop to final insurance.